Transfer: waves and signals

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose equations or diagrams that fit the wave. Explain what the signal tells you.

SUCCESS CRITERIA - I can...

1. Use wave speed and round-trip geometry correctly.
2. Separate frequency, amplitude and direction effects.
3. State the limits of an inference from a signal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your answer. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

round trip: travel from start to a reflector and back

01 Warm-up: bring back an earlier idea

Two targets each have range **500 m** but different radial velocities. **Explain how they can have** equal echo delays yet different Doppler shifts.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $v = f\lambda$; $T = \mathbf{1}/f$; $d_{range} = v\,\Delta t/\mathbf{2}$.
d_{range} is one-way range from round-trip time. Match speed and equation to the wave and medium.

Worked reasoning: An echo time measures range using travel speed, while a frequency shift can indicate radial motion. Different observations answer different questions about the same target.

Your turn: Draw a measurement plan labeling outgoing pulse, return pulse and the time interval.

Sentence starter: My design includes ... . It does what is needed by ... .

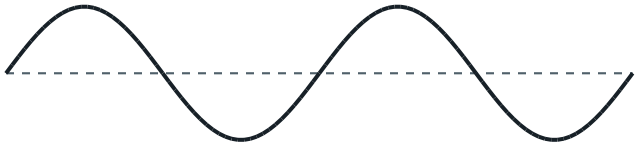


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Sketch a wave** and **label amplitude** and wavelength **0**.
2. **Explain which tells about oscillation** size and which about spatial repetition.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .



Sketch and label your response.

Marzano check - circle: ① ② ③

04 Apply the physics

A radar receives an echo **0.012 s** after transmission. Use $c = \mathbf{3 \times 10^8\ m/s}$ to **find range**. The radar frequency is $\mathbf{3 \times 10^9\ Hz}$; **find wavelength** in vacuum. **Label the timing measurement** and distinguish propagation speed from target motion.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A source gets louder while its frequency remains fixed. Does pitch necessarily rise? Explain using wave properties.

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

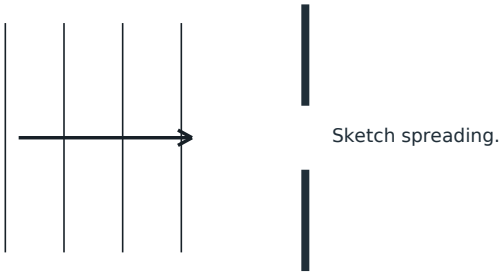
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed wave speed, frequency becomes **4** times as large. **Compare wavelength** and typical diffraction through the same opening.
Label wavelength and opening width; **sketch the emerging wavefronts**.

Word help: diffraction = wave spreading around an edge or through an opening

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

07 Exit check

An ideal blackbody temperature doubles while a sound source frequency doubles. These are separate cases. **Compare each wavelength change under** the facts treated as true for that case.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both ... . They differ in ... because ... .

Case	Quantity doubled	Wavelength comparison
Thermal spectrum	Temperature (K)	____
Sound, fixed speed	Frequency (Hz)	____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A stationary radar measures echoes at **10.0** and **9.6 microseconds** separated by **0.2 s**. **Find both ranges** and average radial velocity; distinguish this from the propagation speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3